

Fall 1998

Problem 1 (Sp86, Sp94, Sp96, Fa98). Let K be a compact subset of $\mathbb{R}^n$ and $\{B_j\}$ a sequence of open balls that covers K . Prove that there is a positive number ε such that each ε -ball centered at a point of K is contained in one of the balls B_j .

Problem 2 (Fa98). Find a function $y(x)$ such that $y^{(4)} + y = 0$ for $x \geq 0$, $y(0) = 0$, $y'(0) = 1$ and $\lim_{x \rightarrow \infty} y(x) = \lim_{x \rightarrow \infty} y'(x) = 0$.

Problem 3 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26). Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \pi \alpha}$$

What restrictions must be placed on α ?

Problem 4 (Fa98, Fa20). Let f be analytic in the closed unit disc, with values $f(-\log 2) = 0$ and $|f(z)| \leq |e^z|$ for all z with $|z| = 1$. How large can $|f(\log 2)|$ be?

Problem 5 (Fa98). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function such that

- the function g defined by $g(x, y) = f(x + y) - f(x) - f(y)$ is *bilinear*,
- for all $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$, $f(tx) = t^2 f(x)$.

Show that there is a linear transformation $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $f(x) = \langle x, Ax \rangle$ where $\langle \cdot, \cdot \rangle$ is the usual inner product on $\mathbb{R}^n$, in other words, f is a quadratic form.

Problem 6 (Fa98). Let A and B be linear transformations on a finite dimensional vector space V . Prove that $\dim \ker(AB) \leq \dim \ker A + \dim \ker B$.

Problem 7 (Fa98). A real symmetric $n \times n$ matrix A is called *positive semidefinite* if $x^T A x \geq 0$ for all $x \in \mathbb{R}^n$. Prove that A is positive semidefinite if and only if $\operatorname{tr} AB \geq 0$ for every real symmetric positive semidefinite $n \times n$ matrix B .

Problem 8 (Fa98). Let R be a finite ring with identity. Let a be an element of R which is not a zero divisor. Show that a is invertible.

Problem 9 (Fa98). Suppose that G is a finite group such that every Sylow subgroup is normal and abelian. Show that G is abelian.

Problem 10 (Fa98). Let f be a real function on $[a, b]$. Assume that f is differentiable and that f' is Riemann integrable. Prove that

$$\int_a^b f'(x) \, dx = f(b) - f(a)$$

Problem 11 (Fa98). Find the minimal value of the areas of hexagons circumscribing the unit circle in $\mathbb{R}^2$.

Note: See also ??.

Problem 12 (Fa98). Let $\varphi(x, y)$ be a function with continuous second order partial derivatives such that

- $\varphi_{xx} + \varphi_{yy} + \varphi_x = 0$ in the punctured plane $\mathbb{R}^2 \setminus \{0\}$,
- $r\varphi_x \rightarrow \frac{x}{2\pi r}$ and $r\varphi_y \rightarrow \frac{y}{2\pi r}$ as $r = \sqrt{x^2 + y^2} \rightarrow 0$.

Let C_R be the circle $x^2 + y^2 = R^2$. Show that the line integral

$$\int_{C_R} e^x (-\varphi_y \, dx + \varphi_x \, dy)$$

is independent of R , and evaluate it.

Problem 13 (Fa98). Let f be an entire function. Define $\Omega = \mathbb{C} \setminus (-\infty, 0]$, the complex plane with the ray $(-\infty, 0]$ removed. Suppose that for all $z \in \Omega$, $|f(z)| \leq |\log z|$, where $\log z$ is the principal branch of the logarithm. What can one conclude about the function f ?

Problem 14 (Fa98). Let $z_1, \dots, z_n$ be distinct complex numbers, and $a_1, \dots, a_n$ nonzero complex numbers such that $S_p = \sum_{j=1}^n a_j z_j^p = 0$ for $p = 0, 1, \dots, m-1$ but $S_m \neq 0$. Here $1 \leq m \leq n-1$. How many zeros does the rational function

$$f(z) = \sum_{j=1}^n \frac{a_j}{z - z_j}$$

have in $\mathbb{C}$? Why is $m \geq n$ impossible?

Problem 15 (Fa85, Sp97, Fa98). Suppose that A and B are endomorphisms of a finite dimensional vector space V over a field $\mathbb{F}$. Prove or disprove the following statements:

1. Every eigenvector of AB is also an eigenvector of BA .
2. Every eigenvalue of AB is also an eigenvalue of BA .

Problem 16 (Fa98). Let B be a 3×3 matrix whose null space is 2-dimensional, and let $\chi(\lambda)$ be the characteristic polynomial of B . For each assertion below, provide either a proof or a counterexample.

1. λ^2 is a factor of $\chi(\lambda)$.
2. The trace of B is an eigenvalue of B .
3. B is diagonalizable.

Problem 17 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 18 (Fa98). Show that the field $\mathbb{Q}(t_1, \dots, t_n)$ of rational functions in n variables over the rational numbers is isomorphic to a subfield of $\mathbb{R}$.